

Spectral properties of polarized synchrotron and thermal dust emission from power spectrum analyses with QUIJOTE, WMAP and Planck data

Pablo de Ferra Chermaz

Supervisor: José Alberto Rubiño Martín

ABSTRACT

ABSTRACT

1 Introduction

2 Data

All data used in this work consist of three independent maps for each frequency band: one intensity map, corresponding to the Stokes parameter I , and two maps describing linear polarization, given by the Stokes parameters Q and U . In this work, we consider polarization data from three experiments: QUIJOTE-MFI, WMAP, and Planck, which are described below.

2.1 Q-U-I Joint Tenerife (QUIJOTE)

The QUIJOTE experiment (Rubiño-Martín et al., 2023) observes the visible sky from the Teide Observatory (latitude $+28.3^\circ$), covering all regions with elevations greater than 30° in both intensity and polarization. The experiment consists of two telescopes equipped with three polarimetric instruments: the Multifrequency Instrument (MFI), operating in the 10–20 GHz range; the Thirty-GHz Instrument (TGI); and the Forty-GHz Instrument (FGI).

In this work we use data from the 11 and 17 GHz bands of the MFI, which have angular resolutions of 55 and 38 arcmin, respectively. The data are obtained from the Legacy Archive for Microwave Background Data Analysis (LAMBDA)¹. The 11 GHz band is particularly useful for characterizing the polarization of low-frequency foregrounds, primarily synchrotron emission (Adak

et al., 2025; Casas et al., 2025) and Anomalous Microwave Emission (AME, Fernández-Torreiro et al., 2023; González-González et al., 2025).

2.2 Wilkinson Microwave Anisotropy Probe (WMAP)

WMAP was a space-based experiment, which observed the total intensity and polarization of the sky, in the frequency range from 23 to 94 GHz, with five frequency bands. In our analysis, we will include the lowest frequency channel of the WMAP dataset, specifically the K-band centered at 23 GHz with angular resolution of FWHM=52.8 arcmin, obtained from the 9-yr data release and downloaded from LAMBDA database².

2.3 Planck satellite

Planck was a space-based experiment consisting of two instruments, the Low Frequency Instrument (LFI) and the High Frequency Instrument (HFI), observing the full-sky in intensity and polarization. This experiment provided 9 frequency channels, covering a wide frequency range from 30 to 857 GHz. In this work, we use the latest data release, focusing on the lowest channel at central frequency 28.4 GHz, downloaded from the Planck Legacy Archive³ (PLA). Additionally, the third public release data (for the same frequency band is used in this work.

¹<https://lambda.gsfc.nasa.gov/product/quijote/index.html>

²<https://lambda.gsfc.nasa.gov/product/wmap/current/index.html>

³<https://pla.esac.esa.int/>

The original resolution of the 30 GHz Planck maps corresponds to an effective beam of FWHM = 32.65 arcmin, being downgraded to the $N_{side} = 512$ ⁴ resolution in order to maintain the same pixelization as the QUIJOTE and WMAP data.

Our analysis employs two complementary datasets: simulated sky maps generated using the Python Sky Model (PySM) library, which incorporates all major Galactic foreground components; and observational data from the QUIJOTE, WMAP, and Planck experiments. All simulated maps are convolved with the appropriate beam window functions for each respective experiment to ensure realistic instrumental effects.

Before fitting the observed power spectra, we subtract the CMB contribution using the best-fit theoretical Λ CDM spectrum from Planck 2018 (Planck Collaboration, 2020). This spectrum is obtained from the full Plik TTTEEE likelihood combined with low-multipole (lowE) and gravitational lensing constraints. For polarization modes, we subtract both C_ℓ^{EE} and C_ℓ^{BB} , where the latter includes only the contribution from B-modes generated by gravitational lensing, since primordial B-modes of inflationary origin have not yet been detected. This CMB subtraction is essential to isolate the Galactic foreground emission (synchrotron and thermal dust) that constitutes the target of this analysis.

CHANGE THE TITLES, USE FULL NAMES

2.4 QUIJOTE

2.5 WMAP

2.6 Planck

2.6.1 Planck LFI

2.6.2 Planck HFI

2.7 Masks

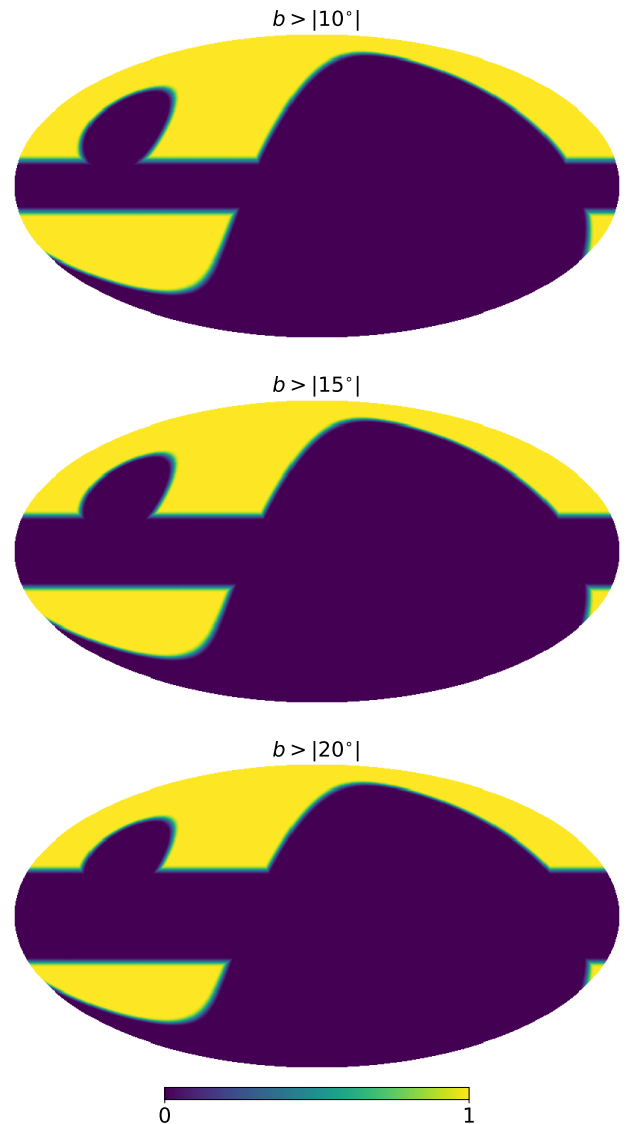


Figure 1: Analysis masks combining the QUIJOTE low-declination survey mask with 10° (upper), 15° (middle) and 20° (lower) cuts in galactic latitude.

⁴<https://healpy.readthedocs.io/en/latest/>

3 Methodology

3.1 Angular power spectra

3.2 Error estimation

Power spectrum uncertainties are estimated through Monte Carlo simulations for both datasets. We compute the standard deviation from 100 realizations each of sky+noise and pure noise maps. The noise characteristics vary by experiment: QUIJOTE and Planck employ the official noise simulations provided by their respective collaborations, incorporating both white noise and correlated $1/f$ noise components. For WMAP, we model the noise as purely white, following standard practice for this dataset.

3.3 Foreground model

The total angular power spectrum between frequencies ν_i and ν_j is modeled as

$$C_\ell(\nu_i, \nu_j) = C_\ell^{\text{sync}}(\nu_i, \nu_j) + C_\ell^{\text{dust}}(\nu_i, \nu_j) + C_\ell^{\text{sd}}(\nu_i, \nu_j), \quad (1)$$

where the synchrotron, dust, and synchrotron-dust cross-correlation terms are

$$C_\ell^{\text{sync}}(\nu_i, \nu_j) = A_s \left(\frac{\ell}{\ell_{\text{ref}}} \right)^{\alpha_s} \left(\frac{\nu_i}{\nu_{\text{ref}}} \right)^{\beta_s} \left(\frac{\nu_j}{\nu_{\text{ref}}} \right)^{\beta_s}, \quad (2)$$

$$C_\ell^{\text{dust}}(\nu_i, \nu_j) = A_d \left(\frac{\ell}{\ell_{\text{ref}}} \right)^{\alpha_d} S_d(\nu_i) S_d(\nu_j), \quad (3)$$

$$C_\ell^{\text{sd}}(\nu_i, \nu_j) = \rho \sqrt{A_s A_d} \left[S_s(\nu_i) S_d(\nu_j) + S_s(\nu_j) S_d(\nu_i) \right] \left(\frac{\ell}{\ell_{\text{ref}}} \right)^{(\alpha_s + \alpha_d)/2}, \quad (4)$$

with $S_s(\nu) = (\nu/\nu_{\text{ref}})^{\beta_s}$ and the modified black-body scaling in Rayleigh-Jeans units

$$S_d(\nu) = \frac{B_\nu(\nu, T_d)}{B_\nu(\nu_0, T_d)} \left(\frac{\nu}{\nu_0} \right)^{\beta_d}, \quad (5)$$

where B_ν is the Planck function. Here, A_s and A_d are the amplitudes of the synchrotron and dust components at the reference multipole $\ell_{\text{ref}} = 80$ and frequencies $\nu_{\text{ref}} = 11.1$ GHz and $\nu_0 = 353.0$ GHz, respectively; α_s and α_d are

the multipole spectral indices; β_s and β_d are the frequency spectral indices; T_d is the dust temperature (fixed at $T_d = 19.6$ K); and ρ is the synchrotron-dust correlation coefficient.

3.4 MCMC fitting

We perform a Markov Chain Monte Carlo (MCMC) fitting procedure using data from QUIJOTE 11 and 17 GHz; WMAP 23, 33, 41, 61, and 94 GHz; and Planck 30, 44, 70, 100, 143, 217, and 353 GHz, covering the multipole range $20 < \ell < 200$. In this study, we use both auto- and cross-spectra between frequency bands. This yields 91 independent spectra from the 14 frequency bands.

3.5 Bayesian Inference

We adopt a Bayesian framework to constrain the model parameters $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$. The posterior distribution is given by

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}), \quad (6)$$

where $\mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})$ is the likelihood and $p(\boldsymbol{\theta})$ is the prior. The likelihood is constructed from the observed power spectra $\hat{C}_\ell(\nu_i, \nu_j)$ and their uncertainties:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i,j,\ell} \frac{[\hat{C}_\ell(\nu_i, \nu_j) - C_\ell(\nu_i, \nu_j; \boldsymbol{\theta})]^2}{\sigma_\ell^2(\nu_i, \nu_j)}. \quad (7)$$

For the power-law model fitting, we adopt uniform (flat) priors for all model parameters within physically motivated bounds:

$$\begin{aligned} A_s &> 0, \\ A_d &> 0, \\ \rho &\in [-1, 1]. \end{aligned}$$

For the bin-to-bin analysis, to regularize the fit and incorporate external information from the literature, we impose a Gaussian prior on the synchrotron frequency spectral index:

$$\beta_s \sim \mathcal{N}(-3.1, 0.30^2), \quad (8)$$

while maintaining uniform priors for the remaining parameters within the bounds specified above.

3.6 Fitting methods

We apply three complementary approaches to the angular power spectra computed from the maps:

Power-law fitting, $\Delta\ell = 10$ We compute the spectra with a bin width $\Delta\ell = 10$ and fit a power-law model for the multipole dependence of the primary foreground components (synchrotron and dust). This fit estimates the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), the multipole spectral indices (α_s, α_d), and the correlation coefficient (ρ) between components.

Power-law fitting, $\Delta\ell = 20$ We repeat the power-law fit using a wider binning, $\Delta\ell = 20$, to test the stability of the fitted parameters with respect to the choice of bin size. Wider bins reduce per-bin noise at the cost of multipole resolution.

Bin-to-bin analysis, $\Delta\ell = 20$ In this approach each multipole bin is fitted independently with $\Delta\ell = 20$, without assuming a global power-law dependence on ℓ . For each bin we only fit the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), and the correlation (ρ); the multipole spectral indices (α_s, α_d) are not constrained in this analysis. This method allows us to identify scale-dependent behaviour and local deviations from the power-law assumption.

4 Results

In the following subsections we first present the results obtained on simulated data and then the results for observational data, applying the three procedures described previously (Section 3.6).

4.1 Power-law fitting

4.2 Bin-to-bin fitting

References

Adak, D., Rubiño-Martín, J. A., Génova-Santos, R. T., Remazeilles, M., Almeida, A., Aryan, K., Ashdown, M., Barreiro, R. B., Bose, U., Cepeda-Arroita, R., Casas, J. M., Fernández-Torreiro, M., Martínez-Gonzalez, E., Poidevin, F., Rebolo, R., and Vielva, P. (2025). QUIJOTE scientific results XIX. New constraints

on the synchrotron spectral index using a semi-blind component separation method. *arXiv eprints*, page arXiv:2510.17761.

Casas, J. M., Bonavera, L., González-Nuevo, J., Rubiño-Martín, J. A., Génova-Santos, R. T., Barreiro, R. B., Cueli, M. M., Crespo, D., Fernández-Fernández, R., and Cano, J. A. (2025). Spatial Variations of Polarized Synchrotron Emission in the QUIJOTE MFI Data. *arXiv eprints*, page arXiv:2510.12860.

Fernández-Torreiro, M., Rubiño-Martín, J. A., López-Caraballo, C. H., Génova-Santos, R. T., Peel, M. W., Guidi, F., Harper, S. E., Artal, E., Ashdown, M., Barreiro, R. B., Casas, F. J., de la Hoz, E., Herranz, D., Hoyland, R., Lasenby, A., Martínez-Gonzalez, E., Piccirillo, L., Poidevin, F., Rebolo, R., Ruiz-Granados, B., Tramonte, D., Vansyngel, F., Vielva, P., and Watson, R. A. (2023). QUIJOTE scientific results - X. Spatial variations of Anomalous Microwave Emission along the Galactic plane. *MNRAS*, 526(1):1343–1366.

González-González, R., Génova-Santos, R. T., Rubiño-Martín, J. A., Peel, M. W., Guidi, F., López-Caraballo, C. H., Fernández-Torreiro, M., Rebolo, R., Hernández-Monteagudo, C., Adak, D., Artal, E., Ashdown, M., Barreiro, R. B., Casas, F. J., Cepeda-Arroita, R., de la Hoz, E., Fasano, A., Herranz, D., Hoyland, R. J., Martínez-Gonzalez, E., Pascual-Cisneros, G., Piccirillo, L., Poidevin, F., Ruiz-Granados, B., Tramonte, D., Vansyngel, F., Vielva, P., and Watson, R. A. (2025). QUIJOTE scientific results: XVIII. New constraints on the polarisation of the anomalous microwave emission in bright Galactic regions: ρ Ophiuchi, Perseus, and W43. *A&A*, 695:A245.

Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A*, 641:A6.

Rubiño-Martín, J. A., Guidi, F., Génova-Santos, R. T., Harper, S. E., Herranz, D., Hoyland, R. J., Lasenby, A. N., Poidevin, F., Rebolo, R., Ruiz-Granados, B., Vansyngel, F., Vielva, P., Watson, R. A., Artal, E., Ashdown, M., Barreiro, R. B., Bilbao-Ahedo, J. D., Casas, F. J., Casaponsa, B., Cepeda-Arroita, R., de la Hoz, E., Dickinson, C., Fernández-Cobos, R., Fernández-Torreiro, M., González-González, R., Hernández-Monteagudo, C.,

Table 1: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and on the synchrotron–dust correlation coefficient ρ , from fits to the EE and BB spectra using the $b > |10^\circ|$ mask.

Data	Mode	$A_s [10^{-3} \mu\text{K}^2]$	α_s	β_s	$A_d [10^{-3} \mu\text{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+Pl	EE	$7.196^{+0.572}_{-0.547}$	$-3.716^{+0.076}_{-0.079}$	$-3.177^{+0.065}_{-0.067}$	$3.178^{+0.019}_{-0.019}$	$-2.542^{+0.006}_{-0.006}$	$1.520^{+0.008}_{-0.008}$	$0.095^{+0.007}_{-0.007}$	7.48
WP+Pl	BB	$2.273^{+0.494}_{-0.437}$	$-3.117^{+0.191}_{-0.202}$	$-3.296^{+0.208}_{-0.232}$	$2.394^{+0.014}_{-0.014}$	$-2.204^{+0.005}_{-0.005}$	$1.555^{+0.008}_{-0.008}$	$0.132^{+0.013}_{-0.013}$	2.28
QJ+WP+Pl	EE	$7.418^{+0.294}_{-0.287}$	$-3.623^{+0.042}_{-0.042}$	$-3.088^{+0.019}_{-0.020}$	$3.181^{+0.019}_{-0.019}$	$-2.542^{+0.006}_{-0.006}$	$1.521^{+0.008}_{-0.007}$	$0.095^{+0.005}_{-0.005}$	5.83
QJ+WP+Pl	BB	$1.703^{+0.206}_{-0.200}$	$-3.291^{+0.122}_{-0.131}$	$-2.960^{+0.061}_{-0.061}$	$2.395^{+0.014}_{-0.014}$	$-2.204^{+0.005}_{-0.005}$	$1.556^{+0.008}_{-0.008}$	$0.114^{+0.011}_{-0.011}$	2.01

Table 2: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and on the synchrotron–dust correlation coefficient ρ , from fits to the EE and BB spectra using the $b > |15^\circ|$ mask.

Data	Mode	$A_s [10^{-3} \mu\text{K}^2]$	α_s	β_s	$A_d [10^{-3} \mu\text{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+Pl	EE	$5.727^{+0.580}_{-0.549}$	$-3.875^{+0.095}_{-0.100}$	$-3.182^{+0.073}_{-0.076}$	$2.066^{+0.017}_{-0.017}$	$-2.514^{+0.008}_{-0.008}$	$1.535^{+0.011}_{-0.011}$	$0.081^{+0.008}_{-0.008}$	6.46
WP+Pl	BB	$2.163^{+0.568}_{-0.498}$	$-3.180^{+0.228}_{-0.254}$	$-3.585^{+0.286}_{-0.327}$	$1.379^{+0.012}_{-0.012}$	$-2.323^{+0.008}_{-0.008}$	$1.539^{+0.012}_{-0.011}$	$0.123^{+0.015}_{-0.015}$	1.63
QJ+WP+Pl	EE	$5.998^{+0.294}_{-0.287}$	$-3.764^{+0.051}_{-0.052}$	$-3.086^{+0.021}_{-0.022}$	$2.066^{+0.017}_{-0.017}$	$-2.514^{+0.008}_{-0.008}$	$1.534^{+0.011}_{-0.011}$	$0.068^{+0.006}_{-0.007}$	5.10
QJ+WP+Pl	BB	$1.363^{+0.218}_{-0.208}$	$-3.405^{+0.157}_{-0.171}$	$-3.003^{+0.072}_{-0.074}$	$1.380^{+0.012}_{-0.012}$	$-2.323^{+0.008}_{-0.008}$	$1.541^{+0.011}_{-0.011}$	$0.118^{+0.013}_{-0.013}$	1.59

López-Caniego, M., López-Caraballo, C., Martínez-González, E., Peel, M. W., Peláez-Santos, A. E., Perrott, Y., Piccirillo, L., Razavi-Ghods, N., Scott, P., Titterton, D., Tramonte, D., and Vignaga, R. (2023). QUI-JOTE scientific results - IV. A northern sky survey in intensity and polarization at 10-20 GHz with the multifrequency instrument. *MNRAS*, 519(3):3383–3431.

Table 3: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and on the synchrotron–dust correlation coefficient ρ , from fits to the EE and BB spectra using the $b > |20^\circ|$ mask.

Data	Mode	$A_s [10^{-3} \mu\text{K}^2]$	α_s	β_s	$A_d [10^{-3} \mu\text{K}^2]$	α_d	β_d	ρ	χ^2_{red}
WP+PI	EE	$6.864^{+0.654}_{-0.628}$	$-3.717^{+0.092}_{-0.098}$	$-3.379^{+0.091}_{-0.096}$	$1.410^{+0.015}_{-0.015}$	$-2.475^{+0.010}_{-0.010}$	$1.460^{+0.013}_{-0.013}$	$0.032^{+0.009}_{-0.009}$	2.64
WP+PI	BB	$1.805^{+0.584}_{-0.492}$	$-3.421^{+0.275}_{-0.310}$	$-3.885^{+0.331}_{-0.376}$	$0.925^{+0.010}_{-0.010}$	$-2.242^{+0.010}_{-0.010}$	$1.534^{+0.016}_{-0.016}$	$0.154^{+0.019}_{-0.018}$	1.56
QJ+WP+PI	EE	$5.908^{+0.292}_{-0.286}$	$-3.726^{+0.052}_{-0.053}$	$-3.129^{+0.024}_{-0.024}$	$1.411^{+0.015}_{-0.015}$	$-2.475^{+0.010}_{-0.010}$	$1.461^{+0.013}_{-0.013}$	$0.022^{+0.007}_{-0.007}$	2.32
QJ+WP+PI	BB	$1.358^{+0.235}_{-0.220}$	$-3.326^{+0.170}_{-0.183}$	$-2.978^{+0.075}_{-0.077}$	$0.926^{+0.011}_{-0.010}$	$-2.242^{+0.010}_{-0.010}$	$1.539^{+0.016}_{-0.016}$	$0.123^{+0.015}_{-0.015}$	1.49

Table 4: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{EE}}$	$A_{\text{dust}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{EE}}$	ρ^{EE}	χ^2_ν
20–39	29.0	296.62 ± 3.82	-3.08 ± 0.02	48.92 ± 0.49	1.48 ± 0.01	0.091 ± 0.007	1.327
40–59	49.0	37.40 ± 1.34	-3.15 ± 0.06	7.81 ± 0.13	1.51 ± 0.02	0.117 ± 0.012	1.186
60–79	69.0	12.79 ± 2.33	-3.30 ± 0.57	4.82 ± 0.21	1.54 ± 0.06	0.046 ± 0.127	1.073
80–99	89.0	6.41 ± 0.88	-3.13 ± 0.34	2.70 ± 0.11	1.53 ± 0.12	0.056 ± 0.022	1.476
100–119	109.0	3.71 ± 0.53	-3.02 ± 0.18	1.53 ± 0.03	1.55 ± 0.03	0.032 ± 0.027	0.968
120–139	129.0	2.85 ± 0.62	-2.58 ± 0.41	0.98 ± 0.03	1.37 ± 0.18	0.158 ± 0.038	1.459
140–159	149.0	2.11 ± 0.62	-2.96 ± 0.38	0.68 ± 0.02	1.47 ± 0.24	0.218 ± 0.056	1.036
160–179	169.0	1.40 ± 0.52	-3.17 ± 0.42	0.47 ± 0.02	1.52 ± 0.07	0.072 ± 0.068	1.037
180–199	189.0	1.30 ± 0.65	-2.91 ± 0.30	0.41 ± 0.02	1.68 ± 0.06	0.112 ± 0.110	1.233
ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{BB}}$	$A_{\text{dust}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{BB}}$	ρ^{BB}	χ^2_ν
20–39	29.0	46.85 ± 4.96	-3.01 ± 0.29	22.86 ± 0.45	1.53 ± 0.08	0.121 ± 0.030	1.339
40–59	49.0	9.53 ± 0.83	-2.97 ± 0.12	6.13 ± 0.09	1.50 ± 0.02	0.111 ± 0.022	1.318
60–79	69.0	1.41 ± 0.54	-3.15 ± 0.40	3.44 ± 0.17	1.57 ± 0.15	0.083 ± 0.065	1.616
80–99	89.0	0.94 ± 0.46	-3.01 ± 0.24	2.05 ± 0.03	1.58 ± 0.02	0.139 ± 0.109	1.429
100–119	109.0	0.81 ± 0.45	-2.72 ± 0.38	1.36 ± 0.03	1.61 ± 0.15	0.111 ± 0.112	1.217
120–139	129.0	1.18 ± 0.44	-3.03 ± 0.25	0.83 ± 0.02	1.56 ± 0.03	0.136 ± 0.081	1.606
140–159	149.0	0.32 ± 0.39	-2.94 ± 0.40	0.54 ± 0.01	1.56 ± 0.08	-0.015 ± 0.170	1.158
160–179	169.0	0.52 ± 0.43	-3.07 ± 0.43	0.44 ± 0.02	1.61 ± 0.08	-0.064 ± 0.164	1.410
180–199	189.0	0.79 ± 0.54	-3.28 ± 0.30	0.40 ± 0.01	1.65 ± 0.04	0.103 ± 0.131	1.376

Table 5: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{EE}}$	$A_{\text{dust}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{EE}}$	ρ^{EE}	χ_{ν}^2
20–39	29.0	275.99 ± 3.85	-3.09 ± 0.02	32.92 ± 0.46	1.44 ± 0.02	0.077 ± 0.008	1.573
40–59	49.0	31.72 ± 1.35	-3.11 ± 0.07	4.87 ± 0.12	1.57 ± 0.03	0.045 ± 0.015	0.975
60–79	69.0	10.72 ± 0.86	-3.33 ± 0.11	2.64 ± 0.06	1.53 ± 0.03	-0.004 ± 0.018	1.078
80–99	89.0	6.00 ± 0.65	-3.15 ± 0.14	1.71 ± 0.04	1.54 ± 0.03	0.079 ± 0.022	1.461
100–119	109.0	3.37 ± 0.61	-2.93 ± 0.35	1.10 ± 0.08	1.53 ± 0.07	0.092 ± 0.029	1.240
120–139	129.0	2.48 ± 0.54	-2.75 ± 0.25	0.60 ± 0.02	1.33 ± 0.04	0.111 ± 0.044	1.400
140–159	149.0	2.04 ± 0.62	-2.93 ± 0.27	0.45 ± 0.01	1.44 ± 0.05	0.143 ± 0.056	1.115
160–179	169.0	1.73 ± 0.57	-3.04 ± 0.28	0.37 ± 0.02	1.60 ± 0.06	0.036 ± 0.060	0.982
180–199	189.0	0.97 ± 0.65	-2.91 ± 0.31	0.30 ± 0.02	1.72 ± 0.09	0.106 ± 0.160	1.241
ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{BB}}$	$A_{\text{dust}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{BB}}$	ρ^{BB}	χ_{ν}^2
20–39	29.0	40.41 ± 2.27	-3.13 ± 0.08	13.26 ± 0.26	1.50 ± 0.02	0.168 ± 0.017	1.284
40–59	49.0	8.52 ± 0.84	-3.00 ± 0.14	4.39 ± 0.08	1.49 ± 0.02	0.045 ± 0.023	1.596
60–79	69.0	1.07 ± 0.54	-3.10 ± 0.27	2.17 ± 0.04	1.58 ± 0.03	0.111 ± 0.097	1.476
80–99	89.0	0.92 ± 0.49	-2.86 ± 0.26	1.11 ± 0.03	1.56 ± 0.04	0.113 ± 0.110	1.276
100–119	109.0	1.04 ± 0.50	-2.84 ± 0.29	0.69 ± 0.02	1.59 ± 0.04	0.039 ± 0.077	1.050
120–139	129.0	1.15 ± 0.44	-3.10 ± 0.28	0.39 ± 0.01	1.57 ± 0.06	0.083 ± 0.067	1.330
140–159	149.0	0.36 ± 0.41	-2.95 ± 0.31	0.33 ± 0.01	1.66 ± 0.07	0.034 ± 0.179	1.116
160–179	169.0	0.62 ± 0.44	-3.18 ± 0.29	0.24 ± 0.01	1.53 ± 0.07	-0.124 ± 0.154	1.297
180–199	189.0	1.50 ± 0.67	-2.99 ± 0.27	0.23 ± 0.01	1.79 ± 0.09	0.029 ± 0.075	1.124

Table 6: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{EE}}$	$A_{\text{dust}}^{\text{EE}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{EE}}$	ρ^{EE}	χ_{ν}^2
20–39	29.0	264.26 ± 4.40	-3.12 ± 0.03	20.39 ± 0.40	1.46 ± 0.02	0.009 ± 0.010	1.518
40–59	49.0	31.67 ± 1.40	-3.13 ± 0.07	4.01 ± 0.10	1.47 ± 0.03	0.065 ± 0.015	1.503
60–79	69.0	10.32 ± 0.82	-3.27 ± 0.12	1.63 ± 0.05	1.45 ± 0.04	-0.088 ± 0.021	0.992
80–99	89.0	5.61 ± 0.66	-3.13 ± 0.15	1.17 ± 0.03	1.49 ± 0.04	0.061 ± 0.024	1.331
100–119	109.0	2.52 ± 0.55	-2.97 ± 0.22	0.81 ± 0.03	1.55 ± 0.05	0.038 ± 0.037	1.109
120–139	129.0	2.21 ± 0.58	-2.78 ± 0.37	0.41 ± 0.02	1.24 ± 0.10	0.161 ± 0.056	1.306
140–159	149.0	2.52 ± 0.62	-2.78 ± 0.24	0.32 ± 0.01	1.43 ± 0.07	0.094 ± 0.050	1.121
160–179	169.0	1.91 ± 0.59	-3.11 ± 0.29	0.27 ± 0.01	1.61 ± 0.08	0.037 ± 0.060	1.063
180–199	189.0	1.00 ± 0.64	-2.90 ± 0.30	0.19 ± 0.01	1.66 ± 0.13	0.114 ± 0.150	1.101
ℓ range	ℓ_{eff}	$A_{\text{sync}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{sync}}^{\text{BB}}$	$A_{\text{dust}}^{\text{BB}} [10^{-3} \mu\text{K}^2]$	$\beta_{\text{dust}}^{\text{BB}}$	ρ^{BB}	χ_{ν}^2
20–39	29.0	37.77 ± 2.31	-3.07 ± 0.09	8.37 ± 0.21	1.52 ± 0.03	0.177 ± 0.021	1.741
40–59	49.0	7.98 ± 0.88	-3.02 ± 0.16	2.93 ± 0.07	1.48 ± 0.03	0.035 ± 0.028	1.258
60–79	69.0	0.99 ± 0.56	-2.98 ± 0.40	1.32 ± 0.04	1.57 ± 0.10	0.171 ± 0.134	1.528
80–99	89.0	0.92 ± 0.51	-2.96 ± 0.40	0.76 ± 0.06	1.57 ± 0.11	0.062 ± 0.097	1.669
100–119	109.0	1.21 ± 0.54	-2.87 ± 0.40	0.47 ± 0.02	1.54 ± 0.10	0.048 ± 0.075	1.045
120–139	129.0	1.13 ± 0.47	-3.12 ± 0.29	0.28 ± 0.01	1.61 ± 0.08	0.094 ± 0.087	1.161
140–159	149.0	0.45 ± 0.47	-2.95 ± 0.31	0.27 ± 0.01	1.79 ± 0.09	0.021 ± 0.184	1.128
160–179	169.0	0.84 ± 0.52	-3.11 ± 0.29	0.17 ± 0.01	1.52 ± 0.10	-0.102 ± 0.133	1.111
180–199	189.0	1.70 ± 0.72	-2.86 ± 0.26	0.15 ± 0.01	1.71 ± 0.12	-0.054 ± 0.087	1.158